

MANTHAN PATEL

Senior Mobile & Embedded Systems Engineer | Real-Time Systems & IoT

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EXECUTIVE SUMMARY

High-impact Senior Mobile & Real-Time Systems Engineer with 5+ years of experience engineering mission-critical, low-latency mobile and hardware-integrated software architectures. Key technical architect behind the Odigo autonomous platform, successfully scaling real-time telemetry and control systems across 100+ production robot deployments in high-density commercial hubs and national infrastructure (Etihad Rail) across the UAE and India. Expert in Flutter, ROS/ROS 2, WebSockets, WebRTC, and Clean Architecture. Recognized for engineering ultra-low latency interfaces (~24ms RTT), leading cross-functional teams, and driving robust enterprise software delivery suited for institutional-grade reliability.

CORE COMPETENCIES & TECHNICAL MASTERY

- **Core Engineering & Architecture:** Clean Architecture, Modular Systems Design, Offline-First Strategy, Concurrency, Performance Profiling, Memory Optimization, High-Availability Systems.
- **Robotics & Hardware Integration:** ROS / ROS 2 Integration, WebSockets, WebRTC Streaming, Serial/UART Protocol, Low-Latency Telemetry Interfaces, Sensor & IoT Data Pipelines.
- **Mobile & Front-End Engineering:** Flutter, Dart, Native Android (Java), iOS, Advanced Reactive UI, Cross-Platform Architecture, State Management (Riverpod, BLOC, Provider).
- **Data, Cloud & Security:** REST APIs (Dio, HTTP), Local Storage (Hive, Sqflite, ObjectBox), Firebase Suite, OCPP (Open Charge Point Protocol), Telnyx WebRTC Encryption.
- **DevOps, Testing & Leadership:** CI/CD Pipelines (Codemagic), Test-Driven Development (TDD), Agile/Scrum Leadership, Architectural Code Reviews, Mentorship.

PROFESSIONAL EXPERIENCE

Senior Flutter & Real-Time Systems Lead | Kody Technolab Limited

Jan 2022 – Present

- **Enterprise Application Architecture:** Architected, built, and deployed 15+ production-grade mobile applications across robotics, fintech, smart mobility, and enterprise communication domains, expanding total career output to 25+ commercial apps.
- **Mission-Critical Systems Integration (Odigo Platform):** Engineered the core control and telemetry interfaces for the Odigo advertising and navigation robot platform, directly enabling over 100+ live autonomous robot deployments across 9 flagship sites in Dubai (e.g., BurJuman) and India (e.g., Jio World Drive, DLF Malls).
- **Ultra-Low Latency Optimization:** Reduced operator-to-robot communication latency to ~24ms round-trip (RTT) with <2ms jitter at 10Hz telemetry by engineering custom protocol bridges uniting ROS/ROS 2, WebSockets, WebRTC, and Serial/UART.
- **National Infrastructure Deployment (Etihad Rail, UAE):** Spearheaded the software implementation for the UAE's first AI robot deployment at a national railway station (Fujairah Station), delivering an offline-resilient multilingual voice assistant and real-time indoor route navigation.
- **Technical Leadership & Governance:** Led and mentored a high-performing team of 3 software engineers. Established strict code review standards, CI/CD automated release pipelines, and sprint planning that streamlined 10+ major production releases.

Flutter Software Engineer | Saubhagyam Web Pvt. Ltd.

Aug 2020 – Jan 2022

- **Full Lifecycle Product Delivery:** Engineered and shipped 10+ cross-platform mobile applications from MVP conceptualization to App Store / Play Store deployment for global B2B and B2C clients in robotics, fintech, and e-learning.
- **Real-Time Data Synchronization:** Integrated complex RESTful API suites and Firebase backends, ensuring seamless offline caching and real-time data sync across distributed user bases.
- **Performance & UI Optimization:** Improved app rendering speeds by 30% through widget tree refactoring, lazy-loading optimizations, and robust state management implementations.

FLAGSHIP ENGINEERING PROJECTS

- **Odigo Autonomous Robot Platform (Robotics / IoT):** Built real-time telemetry dashboards and hardware control layers using ROS/ROS 2, WebRTC, and WebSockets. Operates live in public commercial hubs in Dubai and India.
- **Etihad Rail AI Passenger Assistant (National Transit):** Architected voice-activated NLU assistant with wake-on-tap interaction and dynamic indoor map rendering for passenger guidance at Etihad Rail Fujairah Station.
- **CERO / iCarica / Wuerth EV Network (Fintech & Fleet IoT):** Integrated Open Charge Point Protocol (OCPP) for enterprise EV charging networks across India and MENA, enabling real-time charger status, payment gateway integration, and fleet analytics.
- **Vault Encrypted Communication App (Cybersecurity / Real-Time):** Developed an enterprise-grade secure messaging and calling application utilizing Telnyx WebRTC SDK, end-to-end encryption, and anonymous number masking.
- **Satsang & Karuna Care Platforms (Media & Fintech):** Built high-traffic streaming apps supporting live audio/video streaming, digital subscriptions, and in-app micro-transaction wallet systems.

EDUCATION & LANGUAGES

Bachelor of Computer Applications (BCA) | Gujarat University

2014 – 2017

- **Languages:** English (Professional Working Proficiency), Hindi (Native), Gujarati (Native).